Humidification

Final Lab Report

Unit Operations Lab 4

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1. Executive Summary

The objective of this experiment was to analyze the mass and heat transfer of an air-water system in a wetted-wall humidification column. By determining the experimental mass and heat transfer coefficients, the effect of system variables, such as inlet air temperature and flow rate, was studied to determine if they deviate from theoretical correlations.

Experimental trials were conducted at three air flow rates (2, 4, and 6 scfm) and under three temperature settings (room temperature, low steam, and high steam) using a wetted-wall column equipped with thermocouples and a hygrometer. At each condition, inlet and outlet air temperatures and humidities were measured to calculate the heat transfer coefficient (h), mass transfer coefficient (k_y), number of transfer units (NTU), and height of transfer units (HTU).

At room temperature, k_y increased from approximately 0.0007 kg/s to 0.0025 kg/s as air flow rose from 2 to 6 scfm, while h rose from 1.74 to 3.89 J/k.m^2.s. NTU increased from 0.17 to 0.21, and HTU decreased from 7.63 to 6.44, indicating improved heat and mass transfer due to higher air flow. Under slightly heated conditions (low steam), this effect became more pronounced as k_y increased from 0.0012 kg/s to 0.0313 kg/s, while h rose sharply from 2.98 to 35.23 J/k.m^2.s. NTU increased from 0.29 to 2.59 as HTU fell from 4.55 to 0.51, demonstrating that moderate heating enhanced evaporation. At the highest temperature setting (high steam), k_y remained high but it increased more gradually from 0.0118 kg/s to 0.02702 kg/s, while h rose from 2.72 to 8.11 J/k.m^2.s. NTU increased from 2.23 to 2.92, and HTU remained below 0.60.

These results demonstrate that higher air flow and inlet temperature increase both heat and mass transfer rates by improving contact between the air and water, allowing more water to evaporate. However, at the highest temperature, air reached saturation more quickly, reducing

the driving force for evaporation. Although the observed coefficients were slightly lower than theoretical correlations, the experimental trends show that both mass and heat transfer increase with Reynolds number and temperature, which is consistent with the Lewis relationship. Overall, the experiment validated the coupled nature of heat and mass transfer in humidification processes and the importance of airflow and temperature in improving column performance.

2. Introduction

This experiment aims to investigate the principles of simultaneous heat and mass transfer through the humidification of air in a wetted-wall column. The experiment focuses on understanding the changes in air and water flow rates and the inlet temperature on the total heat and mass transfer coefficients and column performance. Humidification represents one of the most important mass transfer processes, as it illustrates the interactions between thermal and mass transfer operations that occur simultaneously at a gas-liquid interface. These types of operations are crucial in sectors such as air conditioning, drying, cooling towers, and chemical process design, where humidity management and energy efficiency are essential for maintaining product quality and controlling costs. [1]

Humidification occurs when a gas, typically unsaturated, interacts with water vapor, causing water molecules to move from the surface of the liquid into the gas phase until the gas becomes saturated. The process involves simultaneous heat and mass exchange between the water film and the air. The evaporation of water draws energy in the form of latent heat from the surrounding air, decreasing its temperature and changing its humidity. Meanwhile, the air-heated water is cooled to maintain the evaporation. Due to the simultaneous nature of heat and mass transfer and the interdependence between the two, the general rate of humidification is affected by air temperature, air flow, the temperature of the water film, and the disparity between actual and potential humidity. The interaction of these variables determines the efficiency of humidification systems, where dimensionless parameters such as the Reynolds, Sherwood, and Schmidt numbers that determine the characteristics of flow, diffusion, and transfer in the column control the efficiency of the humidification systems.

The experiment is significant because it provides firsthand experience with one of the simplest and most illustrative examples of coupled heat and mass transfer. We were able to identify and interpolate experimental transport coefficients using theoretical correlations by measuring air and water temperatures, humidity, and flow rates. The experiment also uses the Lewis relationship, which relates heat and mass transfer phenomena, demonstrating their interdependence under adiabatic conditions [1]. By understanding such interactions, engineers can design and optimize industrial processes, including evaporative cooling, drying processes, and gas conditioning processes, where heat and mass transfer processes must be optimized together to ensure they achieve effective and cost-effective operation.

The apparatus used in this experiment is an Armfield-type well–wetted column with thermocouples, rotameters, and an air heater, allowing for concurrent air flow and water under controlled conditions. By conducting controlled experiments at various flow rates and air temperatures, the experiment will measure the dependence of heat and mass transfer coefficients on the flow parameters and confirm existing correlations in the form of dimensionless numbers.

Ultimately, the humidification experiment serves as a bridge between theoretical studies and practical applications in industry, focusing on the design concept of effective contactor systems in chemical engineering operations. The experiment provides insight into the interaction of physical principles, including convection, diffusion, and phase change, in a real system by focusing on the simultaneous exchange of heat and mass between the rainbow and the air. The design of industrial equipment, such as cooling towers, air conditioners, dryers, and other gas-liquid contactors, is directly related to these findings, as it is crucial to control humidity and temperature in these devices to ensure energy efficiency and process efficiency. The experiment provides us with practical experience in parameters such as air flow rates through the air and water, temperature gradients, and changes in humidity. it makes us be able to relate abstract transport theory to real behavior, which can be measured.

3. Theory

The process of humidification involves the transfer of heat and mass between a gas and a liquid. In this experiment, a wetted-wall column is used to study the mass and heat transfer that occurs when air flows upward through a tube whose inner wall is coated with a film of water. As the air meets the liquid film, some of the liquid is vaporized, which causes a change in both the humidity and the temperature of the air [2].

The theory of this process lies within the description of transport phenomena, where both heat and mass fluxes are proportional to a driving force. In general, heat flux is proportional to the temperature difference between the phases, while mass flux is proportional to the concentration difference of the transferring component, as expressed in Equations 1 and 2 [3]. The proportionality constants, known as the heat and transfer coefficients, are normally expressed in dimensionless form using correlations that relate them to the Reynolds, Schmidt, and Sherwood numbers (Equation 3). These correlations enable the prediction of transfer coefficients without requiring experimental measurements [1].

In the wetted-wall column, water vapor diffuses from the liquid film into the flowing air stream. As observed in Figure 1, a differential mass balance can be written for a small height of the column, relating the change in vapor concentration to the local driving force for mass transfer (Equation 5). This driving force can be expressed as the difference in mole fraction or mass ratio of water vapor between the interface and the gas. For low concentrations, the rate of diffusion is directly proportional to this difference, and the overall flux can be represented using the mass transfer coefficients [4].

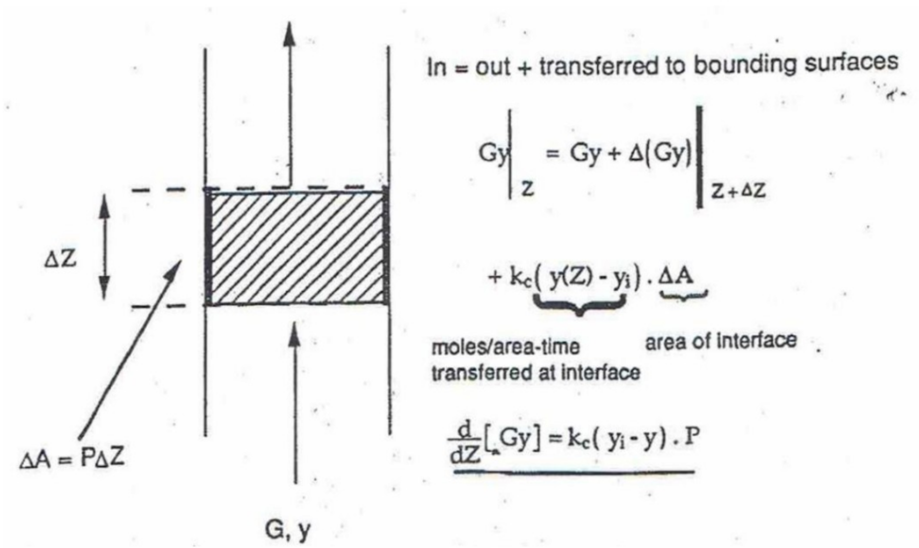

Figure 1: *Differential mass balance for height ΔZ of the column [1]*

A corresponding heat balance can be written over the same differential section of the column. Heat is transferred from the liquid to the gas due to the temperature difference between the two (Equation 10). This heat transfer includes energy that changes the temperature of the air and the energy required to evaporate water into the air. In systems where the vapor concentration is low, the coupling between heat and mass transfer is weak, allowing the two processes to be treated independently. However, for systems with higher mass fluxes, the heat of vaporization must be included, as it can significantly affect the temperature distribution within the column [3].

The experimental data collected, the inlet and outlet air temperatures, and the outlet humidity ratio, are used to calculate the changes in moisture and enthalpy of the air. The humidity ratio is defined as the mass of water vapor per unit mass of dry air (Equation 6). The logarithmic mean difference in humidity ratio is used to evaluate the overall mass transfer coefficient (Equation 12). Similarly, the temperature difference between the gas and the liquid is used to determine the heat transfer coefficient (Equation 13). Both coefficients describe the rate at which mass and energy are transferred from the liquid to the air and are compared to theoretical values to assess model accuracy [5].

The effectiveness of the humidification process is typically expressed in terms of the number of transfer units (NTU) and the height of a transfer unit (HTU). The NTU represents the number of equilibrium stages required to achieve the desired mass of heat transfer, while the HTU represents the efficiency of transfer per unit height [1]. Their product gives the total column height. The NTU is calculated by integrating the differential mass or heat balance equations (Equations 12 and 13), resulting in logarithmic expressions that relate to the inlet and outlet air conditions. These equations quantify how closely the air approaches equilibrium with the liquid.

Overall, the theory of humidification in a wetted-wall column involves both heat and mass transfer between air and water. As air moves through the column, it picks up moisture from the liquid film and changes in temperature and humidity. The rate of these changes depends on how easily heat and water vapor move between the two phases, which is described by the heat and mass transfer coefficients. By measuring the air inlet and outlet temperature and outlet humidity, these coefficients can be calculated and compared to theoretical values to see if they deviate from predicted correlations.

Table 1: Equations used in Humidification Analysis		
Name/Purpose	Equation	Terms Defined
1. Heat Flux: Relates the rate of heat transfer to the temperature difference between two phases	$q=h*DT$	• $q=$¿ heat flux (energy/area-time) • $h=$¿ heat transfer coefficient • $DT=$¿ Characteristic temperature difference
2. Mass Flux: Relates the rate of mass transfer to the concentration difference of the transferring component	$N_A=k_c*DC_A$	• $N_A=$¿ mass flux (moles/area-time) • $h=$¿ heat transfer coefficient • $DC_A=$¿ Characteristic concentration difference

3. <u>Sherwood Correlation:</u> Relates Sherwood, Reynolds, and Schmidt numbers for flow inside a pipe.	$Sh=0.023*Re^{0.83}*Sc^{0.44}$	• $Sc=$¿ Schmidt number • $\Re=$¿ Reynolds number • $Sh=$¿ Sherwood number
4. <u>Sherwood Number Definition:</u> Defines the Sherwood number in terms of the mass transfer coefficient and diffusion coefficient.	$Sh=\frac{k_c L}{D_{AB}}$	• $Sh=$¿ Sherwood number • $k_c=$¿ Mass transfer coefficient for a concentration driving force • $L=$¿ characteristic length scale • $D_{AB}=$¿ diffusion coefficient for air
5. <u>Differential Mass Balance:</u> Change in vapor composition over a small height of the column	$d(Gy)=k_c c(y_i-y)dA$	• $y_i, y=$¿ mole fractions of water at interface and in the gas. • $G=$¿ total gas flowrate (moles/time) • $c=$¿ molar concentration of gas. • $dA=$¿ Area of differential

		column height. • $d(Gy)=¿$ differential change in moles of water in the gas
6. <u>Humidity Ratio:</u> Mass of water vapor per mass of dry air in the gas phase.	$Y'=\frac{\frac{y}{(1-y)}*MW_{H_2O}}{MW_{Air}}$	• $Y'=¿$ humidity ratio • $MW_{H_2O}, MW_{Air}=¿$ molecular weights of water and air, respectively. • $y=¿$the composition of water.
7. <u>Differential Mass Balance with Humidity Ratio:</u> Uses humidity ratio to relate change in humidity to k_y	$G_s' d(Y')=k_y P(Y_i'-Y')dz$	• $G_s'=¿$ mass flowrate of dry air. • $k_y=¿$ mass transfer coefficient in terms of mass ratio driving force. • $Y', Y_i'=¿$ bulk and interface humidity ratio. • $P=¿$ perimeter • $dz=¿$ differential height
8. <u>k_y and k_c relationship:</u> Converts individual to	$k_y=k_c(MW_{Air})c(1-y)_M$	• $(1-y)_M=¿$ log mean

overall coefficient for gas phase		average composition of air. • $k_c = ¿$ Mass transfer coefficient for a concentration driving force. • $c = ¿$ molar concentration of gas. • $MW_{Air} = ¿$ molecular weights of air. • $k_y = ¿$ mass transfer coefficient in terms of mass ratio driving force.
9. Log Mean Average Composition: Defines the log mean average composition of the air over the length of the column.	$(1-y)_M = \frac{y_1 - y_2}{\ln\frac{(1-y_2)}{(1-y_1)}}$	• $(1-y)_M = ¿$ log mean average composition of air. • $y_1, y_2 = ¿$ inlet and outlet vapor fractions
10. Differential Heat Balance: Temperature change of gas due to heat transfer	$d(G C_p T) = h(T_i - T)\,dA$	• $C_p = ¿$ molar heat capacity of the gas. • $T_i = ¿$ temperature at the interface.

		• $T=$¿ gas temperature. • $h=$¿ heat transfer coefficient. • $dA=$¿ Area of differential column height. • $G=$¿ total gas flowrate (moles/time)
11. <u>Humid Heat:</u> Heat capacity of moist air per mass of dry air.	$C_s'=C'_{Air}+Y'C_{H_2O}'$	• $C_s'=$¿ humid heat. • $C_{Air}'=$¿ heat capacity of dry air. • $C_{H_2o}'=$¿ heat capacity of water vapor. • $Y'=$¿ humidity ratio.
12. <u>Integrated Mass Transfer:</u> Relates humidity to k_y and height.	$\ln\frac{Y_1'-Y_i'}{Y_2'-Y_i'}=\frac{k_y Pz}{G_s}$	• $T_i=$¿ temperature at the interface. • $Y_1', Y_2'=$¿ inlet and outlet humidity ratios. • $Y_i'=$¿ interface humidity ratio. • $G_S=$¿ mass flowrate of dry air.

		• $P=$¿ perimeter. • $z=$¿ height.
13. <u>Integrated Heat Transfer:</u> Relates inlet and outlet temperatures to *h* and height.	$\ln\frac{T_1-T_i}{T_2-T_i}=\frac{hPz}{G_s C_s}$	• $T_i=$¿ temperature at the interface. • $T_1, T_2=$¿ inlet and outlet temperatures. • $G_S=$¿ mass flowrate of dry air. • $C_S=$¿ humid heat.

4. Results

Three steady-state runs were completed at three different temperature settings as well as three different inlet air flow rates (2, 4, 6 SCFM), with dry runs taken at each setting to determine the properties of the dry incoming air. The temperature of the heat exchanger was varied by conducting one run at room temperature, followed by a run at low steam temperature, and finally a run at high steam temperature. At each temperature setting, temperature readings were taken at each thermocouple, inlet, and outlet, along with humidity levels measured by the hygrometer.

The collected experimental data were reduced using the humidification relationships provided in the laboratory manual, allowing the mass transfer coefficient (k_y), heat transfer coefficient (h), Sherwood number, NTU, and HTU to be evaluated as a function of Reynolds number, inlet air temperature, and air flow rate. The relationships between these quantities help characterize how interfacial transport changes with turbulence and temperature inside the wetted-wall column.

The heat transfer coefficient can be plotted as a function of the Reynolds number as shown below (Figure 2). It can be noted that heat transfer coefficients seem to have a greater dependence on the Reynolds number than the mass transfer coefficient. At higher flow rates, as the Reynolds number increases, the heat transfer coefficients rise sharply.

Trials conducted at 2 SCFM exhibit mild variation. In contrast, the trials conducted at 6 SCFM show greater variability.

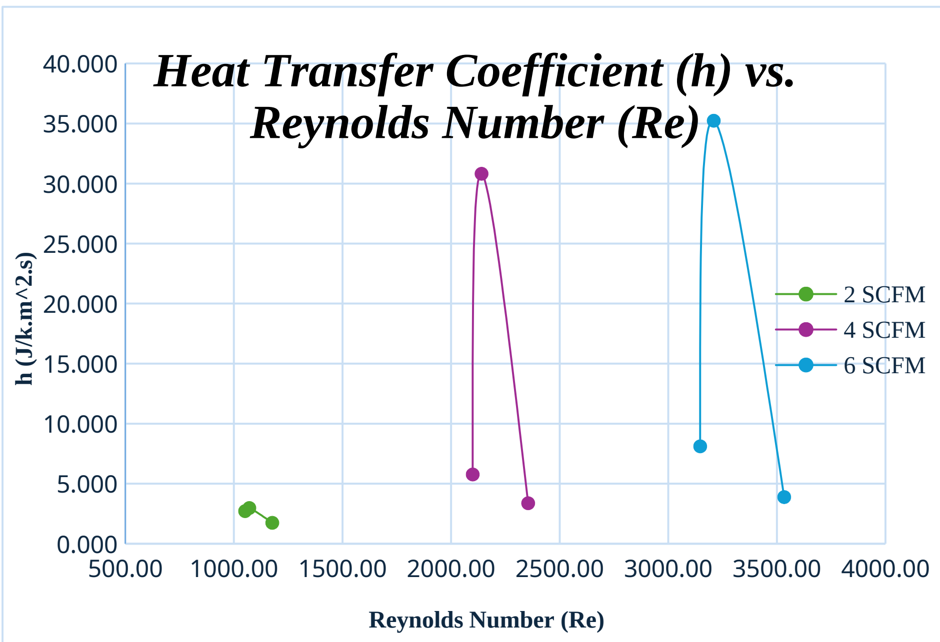

Figure 2: *Heat Transfer Coefficient (h) vs. Reynolds Number (Re)*

Likewise, the mass transfer coefficients are plotted as a function of Reynolds number, where they show a general increase with increasing Reynolds number within a given flow rate. However, the trials conducted at 2 SCFM show a relatively small k_y range.

At higher flow rates, such as 4 and 6 SCFM, the effect of Reynolds number is much more pronounced. As shown in Figure 3, the slopes at a higher Reynolds number is much steeper.

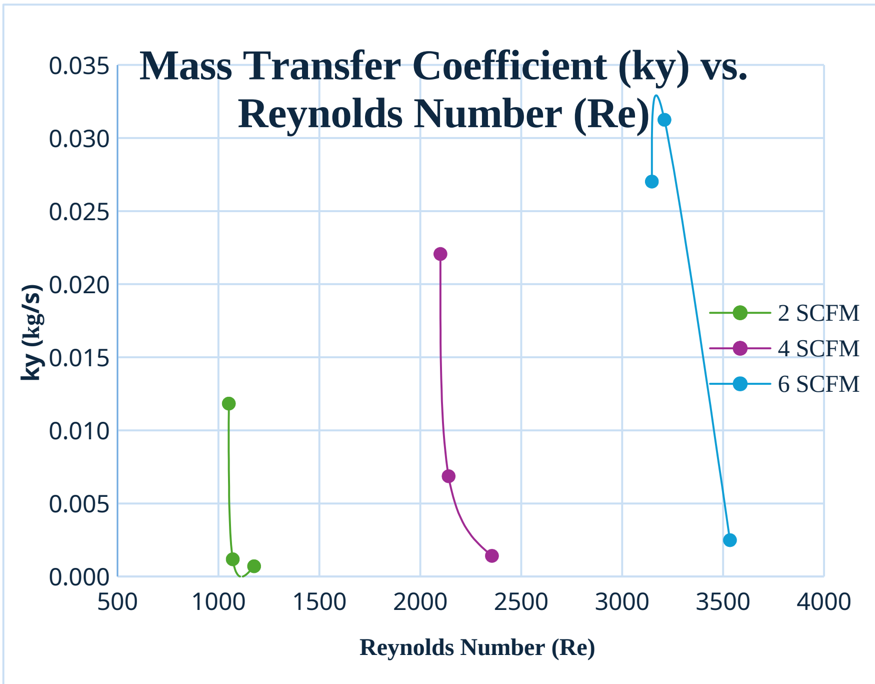

Figure 3: *Mass Transfer Coefficient (ky) vs. Reynolds Number (Re)*

The adjusted Sherwood number can be plotted as a function of the Reynolds number on a log-log plot as follows. (Figure 4) When plotted on a log-log scale and normalized, the experimental Sherwood Numbers fall along the same slope as the empirical correlation given in the manual (1)(*Equation 3*). The alignment of slopes confirms that the dependence on Reynolds number follows the theoretical 0.83 power-law behavior expected for film diffusion in a column.

Small deviations in magnitude between data sets are expected due to inlet humidity fluctuations and imperfect wetting of the column during startup, but do not affect the observed scaling trend.

This agreement confirms that the system operates as expected and that the physical behavior of the interface aligns with humidification theory.

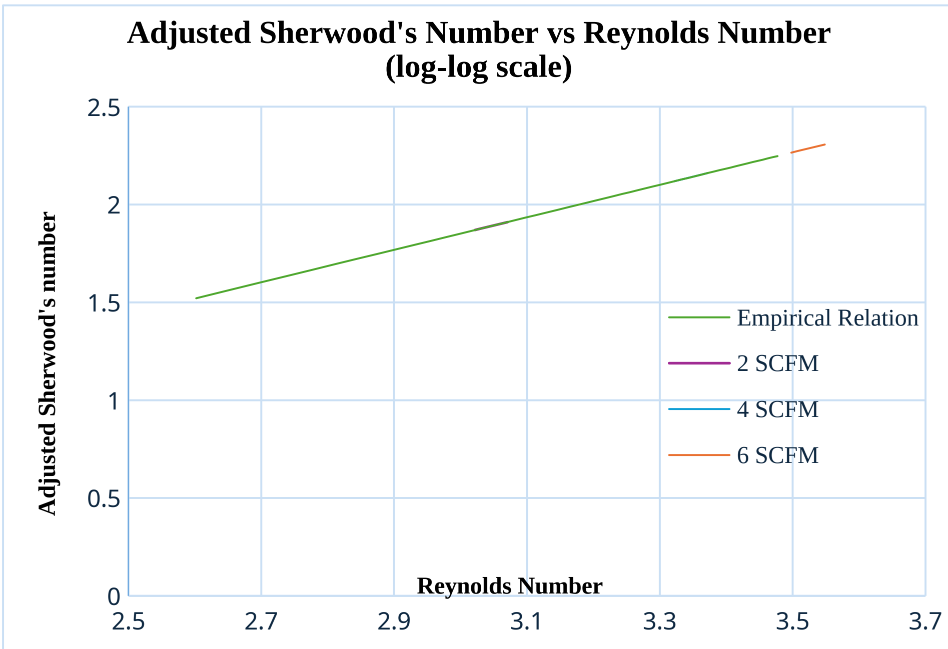

Figure 4: *Adjusted Sherwood Number vs Reynolds Number on a log-log scale*

The data collected can also help determine the number of transfer units (NTU) as a function of inlet air temperatures at the three flow rates (*Figure 5*). NTU had an observable increase with inlet air temperatures, indicating that hotter air temperatures support greater water vapor diffusivity and sustains a larger driving force throughout the column. A notable exception can be made for the 6 SCFM conditions, which show a decline in NTU at higher temperatures. This behavior suggests that transfer is limited at this flow rate.

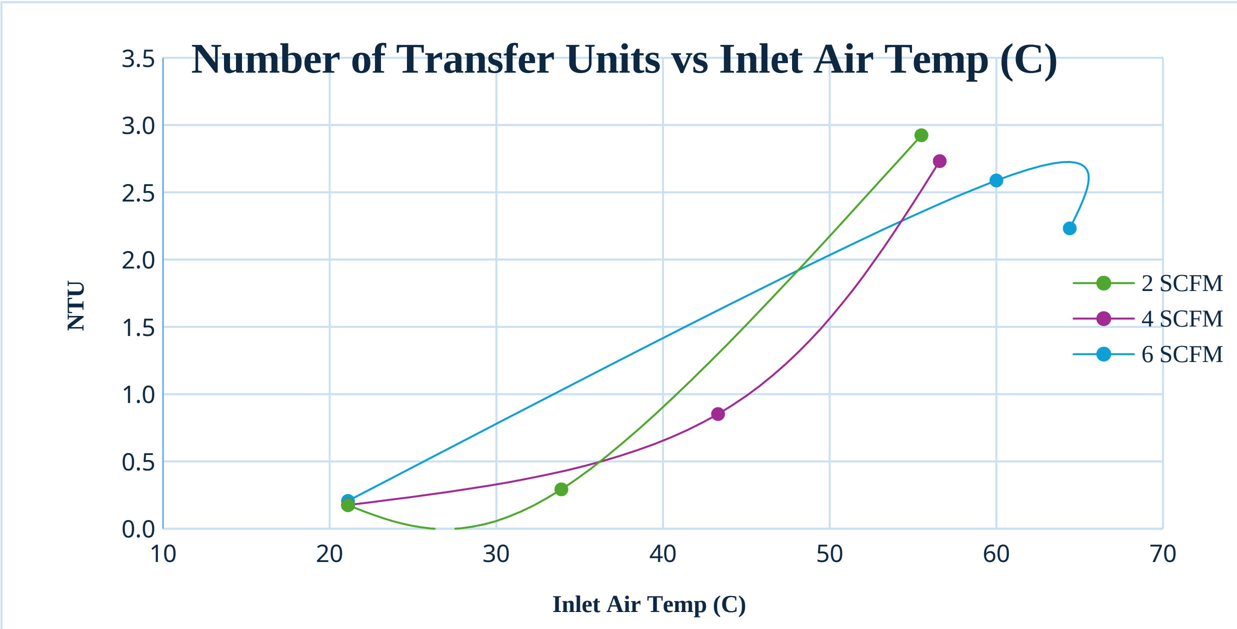

***Figure 5**: Number of Transfer Units vs. Inlet Air Temperature at different air flow rates*

Next, the Lewis relation can be plotted, which predicts that heat and mass transfer should be couples for air-water systems. From the figure shown below (Figure 6) the experimental trends variate significantly from the theoretical line. At the intermediate flow rate, 4 SCFM, the air streams become partially saturated before reaching the top of the column, which sharply reduces the driving force for mass transfer, while convection continues to enhance heat transfer. At the highest flow rate (6SCFM), heat and mass transfer remain more strongly coupled which is why the Lewis ratio gradually returns toward unity.

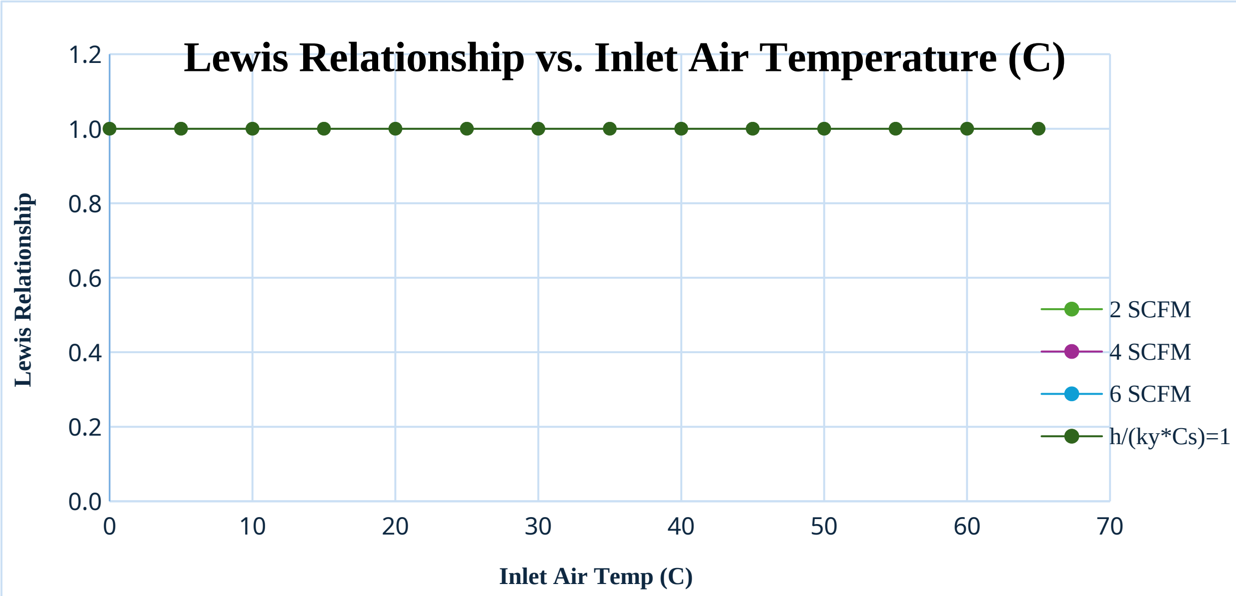

***Figure 6**: Lewis Relationship vs. Inlet Air Temperature (C)*

5. Discussion

The primary objective of this experiment was to evaluate simultaneous heat loss and mass transfer in a wetted-wall humidification column and to determine whether experimentally measured values of the mass transfer coefficient (ky) and heat transfer coefficient (h) behave consistently with established theory. According to the Unit Operation Laboratory Manual, the Sherwood number should scale with Reynolds number following the correlation (1) for diffusion in a circular pipe:

$$Sherwood = 0.023\,\Re^{0.83}\,Sc^{0.44}$$

Therefore, both k_y and h were expected to increase with a higher air flow rate and, therefore, a higher Reynolds number. The dimensionless comparison $\frac{Sh}{Sc^{0.44}}$ plotted against the Reynolds number on a log-log scale should follow a trend line with a slope near 0.83 if the column behaves in a manner similar to the idealized geometry developed in Equation (3) in the manual (1).

Our results partially support this theoretical expectation. As air velocity increased, the experimentally determined transport coefficients generally followed an increasing trend, confirming that interfacial resistance decreases when turbulence begins to thin the mass and heat boundary layers. Specifically, k_y increased from approximately 0.001 kg/s at low flow rates to over 0.03 kg/s at the highest flow conditions as the Reynolds number increased from roughly 1000 to 3500. The corresponding Sherwood number correlation demonstrated a similar increasing relationship, following the theoretical slope direction, although it consistently fell below the predicted curve. Likewise, the heat transfer coefficient increased from approximately 1.7 to over 35 J/(m²·s·K) under the same operational change, indicating that the enhancement mechanisms for heat transfer are strongly coupled, as predicted by the Lewis relationship.

$$\frac{h}{\left(k_y C_S'\right)} \approx 1$$

Our experimentally tested Lewis ratio was generally between 0.2 and 4.3, which shows qualitative consistency with theory, but quantitative deviation due to experimental non-idealities.

The observed trends in Sherwood number versus Reynolds number clearly demonstrate that increasing air velocity enhances convective mass transfer. As the plots show, each jump in air flow rate shifted the data upward, reflecting lower resistance to water vapor diffusion in faster-moving air. Similarly, in the plots of h versus Reynolds number, the nearly monotonic rise in heat transfer coefficient indicates reduced thermal boundary layer thickness at higher flow rates, leading to more efficient energy exchange at the interface.

The NTU vs. temperature plot further supports this behavior. NTU increased dramatically as the inlet air temperature increased because the saturation pressure difference between the interface and bulk air became much larger, directly amplifying the driving force for mass transfer. In contrast, HTU exhibited a clear decreasing trend with temperature, indicating that the same change in humidity required a lower contact height. Together, these trends illustrate that higher temperatures not only enhance transport rates but also improve equipment performance by reducing required column size.

The experiment also examined the effect of increasing the inlet air temperature. Theory dictates that a higher inlet temperature raises the saturation vapor pressure at the interface, resulting in a larger humidity driving force and, therefore, a higher mass flux into the gas. This was generally observed in the data: when the air was heated to high-temperature conditions of 55–64°C the outlet humidity increased to as high as ~66% RH and k_Y increased accordingly to values exceeding 0.02–0.03 kg/s. In addition to aligning with theory, this result supports mass and energy balance expressions such as Eqns. 12 and 13, which directly express temperature and humidity changes as functions of transport resistances in a counter-current humidification system.

Additionally, the adjusted Sherwood number vs. Reynolds number (log-log) plot showed that data maintained a similar slope across all flow rates, confirming that mass transfer still behaves according to boundary-layer theory, even if the absolute magnitude does not meet ideal values. This indicates that while non-idealities shifted the data downward, the core physics governing humidification remained valid throughout the experiment.

Although the observed trends align with expectations, the magnitudes of the transport coefficients were consistently lower than those predicted by the literature-based correlation, especially at higher Reynolds numbers. This discrepancy can be explained through practical limitations of the wetted-wall column. Some heat leakage occurred through the Plexiglas walls despite the assumption of adiabatic operation, decreasing the effective driving force $(T_i - T)$and lowering the calculated h. Additionally, the water film may not have evenly coated the column surface at lower flow rates, thereby reducing the contact area and limiting mass transfer relative to the assumed geometric conditions. The manual emphasizes that proper full wetting is essential to match theoretical predictions, and any partial surface exposure weakens mass transfer performance. At high humidity, errors in wet-bulb measurement increased the uncertainty in Y', causing a larger deviation in the calculated $k_Y$and Sherwood numbers. Despite these limitations, the consistency of the trends supports the reliability of the underlying experimental methodology.

Certain assumptions, such as dilute vapor concentrations and negligible coupled heat-of-transfer effects on conductivity, were less reliable during the heated runs when the outlet humidity became high. Under those conditions, the simplifications behind Equation (3) become less valid because mass flux is no longer small, and diffusion cannot be assumed equimolar. Nevertheless, the data remained useful because deviations from ideality were systematic and explainable. They do not prevent conclusions about performance behavior, but they should be acknowledged if these values were to be used for equipment design.

Overall, the experiment successfully demonstrated that both mass and heat transfer increase with the Reynolds number and that air heating enhances vapor absorption, thereby validating the core theory of simultaneous transport in humidification systems. However, quantitative disagreement with the theoretical correlation suggests improvements could be made. Instrumentation upgrades, particularly in terms of humidity measurement accuracy and improved thermal insulation, would directly address identified error sources. Additionally, increasing water flow rates to ensure full surface wetting would more closely achieve the intended operating design outlined in the lab manual. A more accurate humidity sensor, better insulation to enforce adiabatic conditions, and a mechanism to visually confirm film wetting would reduce systematic error and increase agreement with established models. Future work could include additional runs at very high Reynolds numbers to evaluate whether the correlation slope converges to the theoretical 0.83 asymptotically, as well as testing different column diameters or surface treatments to directly assess the influence of geometry on transfer coefficients.

In summary, while the exact magnitudes of the transfer coefficients differed from ideal theoretical predictions, the results demonstrated all expected qualitative trends, confirmed the importance of boundary-layer reduction in enhancing transport, and reinforced the coupling of heat and mass transfer in humidification operations. Remaining discrepancies can be attributed to well-characterized experimental limitations rather than flaws in the governing transport theory.

6. References

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7. Appendix

a. Appendix I: Data Tabulation/Graphs

Recorded Raw Data During Trials:

Table 2: Room Temp Trials

Trial	Air Flow Rate (SCFM)	Water Flow Rate(G.P.M)	Inlet Air Temp (°C)	T1 (°C)	T2 (°C)	T3 (°C)	T4 (°C)	T5 (°C)	T6 (°C)	T7 (°F)	Humidity (%)
1 -Dry	2	0	17.8	21.5	21.8	20.9	21.6	21.7	21.6	70.5	6.8
2-Wet	2	0.2282	21.1	22.1	22.9	24.7	24.3	24.3	24.4	72.5	77.5
3-Wet	4	0.2282	21.1	22.7	23.8	26.4	25.8	25.8	25.8	73.9	77.5
4-Wet	6	0.2282	21.1	23.1	24	26.9	26.3	26.2	26.1	74.8	77.5

Table 3: Low Steam Trials

Trial	Air Flow Rate	Water Flow Rate(G.P.	Inlet Air Tem	T1 (°C)	T2 (°C)	T3 (°C)	T4 (°C)	T5 (°C)	T6 (°C)	T7 (°F)	Humidity (%)

	(SCFM)	M)	p (°C)								
1-Dry	2	0	58	31	40.5	27.4	27.9	31.1	24.8	98.6	5.5
2-Wet	2	0.2282	33.9	23.8	26	28.5	27.9	28	27.9	77	84.1
3-Wet	4	0.2282	43.3	24.6	28.5	29	28.4	28.4	28.3	80.6	71.4
4-Wet	6	0.2282	60	29.7	37.8	29.6	29	29	29	95.1	44.2

<u>Table 4: Low Steam Trials</u>

Trial	Air Flow Rate (SCFM)	Water Flow Rate(G.P.M)	Inlet Air Temp (°C)	T1 (°C)	T2 (°C)	T3 (°C)	T4 (°C)	T5 (°C)	T6 (°C)	T7 (°F)	Humidity (%)
1-Dry	2	69	41.3	51.3	45.2	45.6	46.2	36.4	103.2	4.2	69
2-Wet	2	55.5	21.5	35	30.7	30	30.2	30.3	91	66.1	55.5
3-Wet	4	56.6	30	36.8	31.2	30.6	30.7	30.8	93	57.3	56.6
4-	6	64.4	32.6	41.	31.4	30.	30.	30.	100.	43.5	64.4

Wet				3		8	8	9	7		

Mass Fraction Tables:

Table 5: Mass Fractions at Room Temp											
Trial	Ti(°C)	T1 (°C)	T7 (°F)	Humidity (%)	Yi (kg water/kgair)	Y1 (kg water/kgair)	Y2 (kg water/kgair)	Gs (kg/s)	ky (kg/s)	NTU	HTU
2-Wet	24.43	22.1	72.5	77.5	0.0147	0.0127	0.0130	0.0011	0.0007	0.174	7.628
3-Wet	26	22.7	73.9	77.5	0.0161	0.0132	0.0137	0.0023	0.0014	0.175	7.568
4-Wet	26.46	23.1	74.8	77.5	0.0166	0.0135	0.0141	0.0034	0.0024	0.206	6.443

Table 6:Mass Fractions at Low Temp Steam											
Trial	Ti(°C)	T1 (°C)	T7 (°F)	Humidity (%)	Yi (kg water/kgair)	Y1 (kg water/kgair)	Y2 (kg water/kgair)	Gs (kg/s)	ky (kg/s)	NTU	HTU
2-Wet	31	23.8	77	84.1	0.0199	0.0153	0.0165	0.0011	0.0011	0.2921	4.547
3-Wet	0	24.6	80.6	71.4	0.0173	0.0136	0.0157	0.0023	0.0068	0.8518	1.559
4-Wet	0	29.7	95.1	65.3	0.0110	0.0114	0.0154	0.0034	0.0312	2.5879	0.513

Table 7: Mass Fractions at High Temp Steam											
Trial	Ti(°C)	T1 (°C)	T7 (°F)	Humidity (%)	Yi (kg water/kgair)	Y1 (kg water/kgair)	Y2 (kg water/kgair)	Gs (kg/s)	ky (kg/s)	NTU	HTU
2-Wet	0	21.5	91	66.1	0.0177	0.0681	0.0204	0.0011	0.0118	2.9242	0.454
3-Wet	0	30	93	57.3	0.0158	0.06205	0.0188	0.0023	0.0220	2.7319	0.486
4-	0	32.6	100.7	43.5	0.01	0.06	0.018	0.00	0.02	2.23	0.59

Wet					20	772	0	34	70	19	5

Further Calculations:

Table 8: Calculations at Room Temperature

Trial	Cs (J/kg.K)	h (J/k.m^2.s)	Lewis Relationship	Superficial V (m/s)	Reynolds number	Diffusion coefficient (m^2/s)	Schimdt Number	Sherwood Number
2-Wet	1032.4	1.740512	2.411456948	0.2578346	1176.72	2.48E-05	0.60654	65.28965
3-Wet	1035.12	3.372775	2.31040053	0.5160038	2354.96	2.51E-05	0.60099	115.6584
4-Wet	1035.98	3.886043	1.509009341	0.7743354	3533.95	2.51E-05	0.59936	161.795

Table 9: Calculations at Low Temperature Steam

Trial	Cs (J/kg.K)	h (J/k.m^2.s)	Lewis Relationship	Superficial V (m/s)	Reynolds number	Diffusion coefficient (m^2/s)	Schimdt Number	Sherwood Number
2-Wet	1042.11	2.975764	2.426081566	0.258739	1070.97	2.54E-05	0.65446	62.43564
3-Wet	1037.34	30.81559	4.330498533	0.517089	2140.34	2.54E-05	0.65269	110.7895
4-Wet	1025.63	35.23432	1.099272929	0.7753814	3209.46	2.55E-05	0.65042	154.8364

Table 10: Calculations at High Temperature Steam

Trial	Cs (J/kg.K)	h (J/k.m^2.s)	Lewis Relationship	Superficial V (m/s)	Reynolds number	Diffusion coefficient (m^2/s)	Schimdt Number	Sherwood Number
2-Wet	1038.01	2.715396	0.221127153	0.2597692	1051.56	2.57E-05	0.66085	61.75868
3-Wet	1034.47	5.766966	0.252618795	0.5186911	2099.7	2.58E-05	0.65883	109.4907
4-	1027.	8.1113	0.292112996	0.77743	3147.1	2.58E-0	0.658	153.133

Wet	54	01		45	1	5	19	7

Lewis Relationships:

Table 11:

Calculations at Flow Rate 2

Trial	Lewis relationship	Inlet air Temp (°C)
4	2.41145695	21.1
10	1.50900934	33.9
16	0.22112715	55.5

Table 12:

Calculations at Flow Rate 4

Trial	Lewis relationship	Inlet air Temp (°C)
5	2.3104005	21.1
11	4.3304985	43.3
17	0.2526188	56.6

Table 13:

Calculations at Flow Rate 6

Trial	Lewis relationship	Inlet air Temp (°C)
6	1.5090093	21.1
12	1.0992729	60
18	0.292113	64.4

Log-log plot calculations:

Table 14

Sherwood's Number	**Reynolds Number**	**log (Sherwood)**	**log (Reynolds)**
33.2229069	400	1.52143763	2.602059991
36.6348335	450	1.56389422	2.653212514
39.9827756	500	1.60187294	2.698970004
43.2741821	550	1.63622887	2.740362689
46.5150373	600	1.66759337	2.77815125
49.7102471	650	1.69644592	2.812913357
52.8639012	700	1.72315921	2.84509804
55.9794574	750	1.74802868	2.875061263
59.0598752	800	1.77129253	2.903089987
62.107714	850	1.79314554	2.929418926
65.1252073	900	1.81374912	2.954242509
68.1143205	950	1.83323843	2.977723605
71.0767949	1000	1.85172784	3
74.0141838	1050	1.86931495	3.021189299
76.9278801	1100	1.88608376	3.041392685
79.8191399	1150	1.90210704	3.06069784
82.689101	1200	1.91744827	3.079181246
85.5387987	1250	1.93216315	3.096910013
88.369178	1300	1.9463008	3.11394335

6		2	2
91.1811077	1350	1.95990486	3.130333768
93.9753833	1400	1.97301411	3.146128036
96.7527411	1450	1.98566328	3.161368002
99.513862	1500	1.99788358	3.176091259
102.259378	1550	2.00970315	3.190331698
104.989876	1600	2.02114742	3.204119983
107.705904	1650	2.03223951	3.217483944
110.407974	1700	2.04300044	3.230448921
113.096565	1750	2.05344942	3.243038049
115.772128	1800	2.06360402	3.255272505
118.435084	1850	2.07348037	3.267171728
121.08583	1900	2.08309332	3.278753601
123.724744	1950	2.09245656	3.290034611
126.352178	2000	2.10158273	3.301029996
128.968468	2050	2.11048354	3.311753861
131.573931	2100	2.11916985	3.322219295
134.168869	2150	2.12765176	3.33243846
136.753566	2200	2.13593866	3.342422681
139.328296	2250	2.14403933	3.352182518
141.893317	2300	2.15196194	3.361727836
144.448874	2350	2.15971416	3.371067862
146.995204	2400	2.16730317	3.380211242

149.532532	2450	2.17473569	3.389166084
152.06107	2500	2.18201804	3.397940009
154.581026	2550	2.18915619	3.40654018
157.092595	2600	2.19615571	3.414973348
159.595967	2650	2.20302191	3.423245874
162.09132	2700	2.20975976	3.431363764
164.578831	2750	2.21637397	3.439332694
167.058663	2800	2.222869	3.447158031
169.530979	2850	2.22924907	3.45484486
171.995932	2900	2.23551817	3.462397998
174.45367	2950	2.24168011	3.469822016
176.904336	3000	2.24773848	3.477121255

b. Appendix II: Error Analysis

This humidification experiment was affected by both random and systematic errors that influenced the accuracy and precision of the calculated heat and mass transfer coefficients. This lab is designed under several simplifying assumptions, including steady state operation, negligible external heat loss, and a fully wet column wall. Yet the reality of operating a wetted-wall column made these conditions difficult to achieve (1). As a result, deviations from ideal performance are expected. One major procedural source of error was not waiting long enough to reach a true steady state before recording data. If measurements were taken while temperatures and humidity were still changing with time, the gradients driving heat and mass transfer would not represent the final stabilized values assumed in the calculations. This would directly affect computed NTU, HTU, h, and k_Y, since these values depend on accurate steady-state inlet and outlet air properties. Runs with insufficient stabilization time likely produced artificially high or low vapor content differences, leading to inconsistencies in the calculated coefficients.

Random errors primarily arose from the temperature and humidity measurements used to determine the driving forces for transport. Since humidification relies on small differences in temperature and vapor content between the inlet and outlet air streams, even slight fluctuations were amplified in the calculated values of h, $k_{y,}$ and NTU. As indicated by the manual, both transport coefficients depend strongly on these measured values, and a small drift in the wet-bulb thermometer, electric hygrometer, wall thermocouples, and rotameters would scatter these results. Additionally, water flowing down the Plexiglas tube may not have perfectly coated the entire internal surface, which would lead to variation in temperature and mass transfer. All these random variations contribute to the spread observed in plots of heat and mass transfer versus Reynolds number, helping to explain why repeated runs under identical settings did not yield perfect coefficients.

Systematic errors contributed away from the theoretical behavior. Although the setup is assumed to operate adiabatically, some heat leaks through the transparent column, and recirculation lines are unavoidable, which could lead to lower temperature driving forces than assumed in theoretical correlations (1). The interfacial temperature, needed to determine water saturation pressure and equilibrium humidity, was not measured directly but estimated from the wall thermocouple readings. Any discrepancies in this estimate inevitably affect the calculated water vapor content in the gas phase and, therefore, impact both the mass transfer calculations and NTU values. Similarly, the humidity sensor is less precise when the outlet air approaches saturation, and the assumption used in the data reduction that the heat capacity of the humid air remains constant conflicts with the manual's note that it changes as water vapor content rises (1). Finally, the theoretical correlation used for comparison, Equation (3), strictly applies only when the diffusing component is dilute and the mass flux is low. However, the outlet humidities in several runs were relatively high, meaning that the required assumption was not fully satisfied. Not achieving a steady state also introduces systematic bias because changing water temperature or humidity over time means the calculated coefficients represent transitional behavior rather than true equilibrium operation. Early-time data may underestimate column efficiency, as the driving force for mass transfer is still evolving toward its final, stabilized value.

The combined effect of these systematic influences was that the experimental Sherwood numbers and heat transfer coefficients generally fell below ideal predictions. Nonetheless, despite these uncertainties, the experiment successfully reproduced the correct qualitative behavior. Higher air flow rates increased both heat transfer and mass transfer by strengthening the driving forces and reducing boundary layer resistance, and heating the incoming air led to greater water uptake because the saturated vapor pressure increased. In other words, while some quantitative disagreement with the correlation was observed, the trends aligned with humidification theory and demonstrated the expected physics.

Overall, the limitations of the equipment and measurement techniques resulted in both scatter in the collected data and small, consistent shifts away from theoretical values. However, the results remain trustworthy in establishing the principles of simultaneous heat and mass transfer and validate why dimensionless correlations are useful tools for designing and scaling up humidification units. Future iterations of this experiment should allow more time for temperature and humidity values to stabilize before measurements are taken, ensuring that all results accurately reflect steady-state conditions.

c. Appendix III: Sample Calculations

The sample calculation below shows how the heat and mass transfer coefficients calculated during the experiment of humidification were obtained with the help of the data provided in Trial 4 (wet run)

Convert the air flow to mass flow

$$2\,SCFM = 2 \times 0.0283\,m^3/min = 0.0566\,m^3/min = 9.43 \times 10 - 4\,m^3/s$$

At 25 °C and 1 atm, density of air ≈ 1.18 kg/m³

$$GS\square' = 1.18\left(9.43 \times 10^{-4}\right) = 1.11 \times 10^{-3}\,kg\,air/s$$

Assume humid-air heat capacity $CS\square' = 1.05 \times 10^3\,Jk\,g^{-1}\,{}^\circ C^{-1}$.

1. Mass transfer coefficient (k_Y)

From Equation 12 in the manual:

$$\ln\left(\frac{Yi\square' - Y\,2\square'}{Yi\square' - Y\,1}\right) = \frac{kY\ \ Pz}{GS\square'}$$

$$\ln\left(\frac{0.0198 - 0.0165}{0.0198 - 0.0099}\right) = \ln(3.0) = 1.099$$

$$k_Y = \frac{GS\square'(1.099)}{Pz} = \frac{\left(1.11 \times 10^{-3}\right)(1.099)}{(0.160)(1.22)} = 6.25 \times 10 - 3\,kg\,H^2O\,m^{-2}\,s^{-1}\left(\Delta Y'\right)^{-1}$$

2. Heat Transfer Coefficient (h)

From Equation 13:

$$\ln\left(\frac{Ti - T_1}{Ti - T_2}\right) = \frac{hPz}{GS\square' CS\square'}$$

$$\ln\left(\frac{24.7 - 21.1}{24.7 - 22.9}\right) = \ln(2.0) = 0.693$$

$$h = \frac{GS\square' CS\square' (0.693)}{Pz} = \frac{(1.11\times10^{-3})(1.05\times10^{3})(0.693)}{(0.160)(1.22)} = 3.9\times10^{0}\, W\, m^{-2}\, {}^\circ C^{-1}$$

3. Number and height of Transfer units

$$NTU = \ln\left(\frac{Yi\square' - Y_1'}{Yi\square' - Y_2'}\right) = 1.099$$

$$HTU = \frac{z}{NTU} = \frac{1.22}{1.099} = 1.11\, m$$

4. Lewis's Relationship

$$\frac{h}{k_Y CS\square'} = \frac{3.9}{(6.25\times10-3)(1.05\times10^{3})} = 0.59 \approx 1$$